



Scuba diving

Our understanding of the oceans has been helped a lot by our ability to dive beneath the surface and see the oceanic world for ourselves. This was made practical with the development of scuba diving gear in the mid-20th century. By providing divers with the means to swim freely to depths of 100 ft (30 m) or more, it opened a new era of underwater exploration.



HARD HAT DIVERS

In the past, a diver was equipped with a waterproof suit sealed to a metal helmet, which was filled with air supplied through a rubber tube from a boat. Weighted boots stopped divers from floating to the surface, but forced them to walk on the seabed. The system worked well enough for static jobs such as inspecting harbor walls, but it was useless for exploring the underwater world.

Long fins on the diver's feet increase swimming speed

Buoyancy jacket can be inflated with air to adjust buoyancy underwater

Diver draws in air through a tube from the cylinder

AQUA-LUNG

The letters of the word scuba stand for self-contained underwater breathing apparatus—a system that uses cylinders of compressed air carried on the diver's back. It was invented in 1942 by French ocean explorer Jacques Cousteau, and was originally called the Aqua-Lung. Unlike the earlier hard hat system, it enables a diver to swim in open water, and does not require a special suit; some scuba divers wear suits, but only to keep warm and to protect themselves from stinging organisms in the water.